14 Perimeter and Area

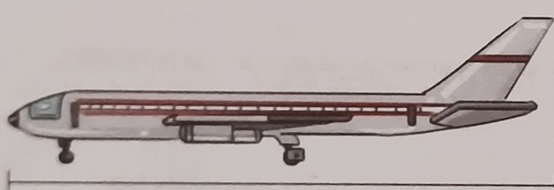
Tune In

Go to MyInspire for

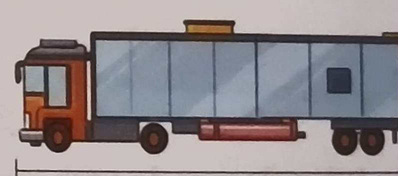
There are lots of different modes of transport. These help people move on land, in the air or water. An aeroplane is the fastest mode of transportation.

Now read the following clues and find the lengths of the given transports.

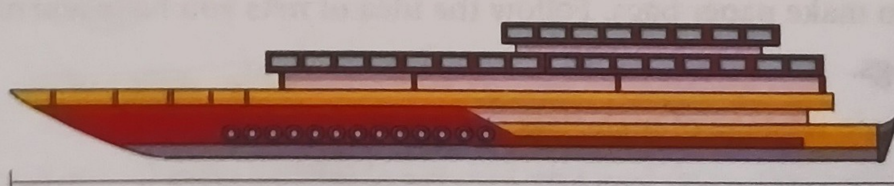
- The length of the truck is about three times 6 m.
- The length of the aeroplane is 22 m more than the length of the truck
- The yacht is the longest mode of transport among the following. The difference between the length of the aeroplane and the yacht is 79 m.



Length = ________ m



Length = ________ m



Length = ________ m

The total length of all the above transports is ____________.

PERIMETER

Rohan wants to paste a green ribbon all along the sides of painting he had made for his sister.

To find the length of the ribbon needed, he measured the sides of the picture. He found that length of the picture is 35 cm and its breadth is 25 cm.

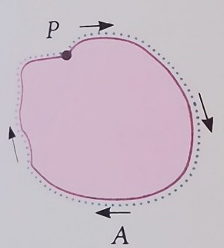
So, length of the ribbon needed is
35 cm + 25 cm + 35 cm + 25 cm = 120 cm.

Here the total length of the ribbon is actually the perimeter of the frame. So here 120 cm is the perimeter of the picture.

The length of the boundary of a closed figure is called its perimeter. In other words, perimeter is the sum of all the sides of a polygon.

Perimeter of Curved Shapes

In the given figure, from any point *P* if we move along the boundary, we again reach the point *P*. So, it is a closed figure.

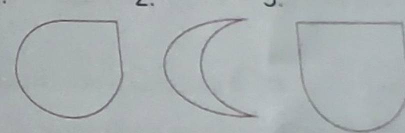


You can use a string to measure the perimeter. Starting from any point, place the string once along the boundary.
The perimeter can then be found by measuring the length of the string using a ruler.

Let's Try! 1

Find the perimeter of the following shapes using a string and a ruler.

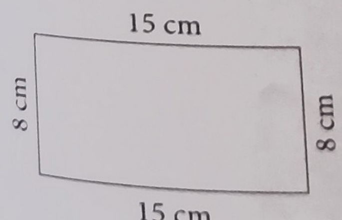
1. 2. 3.



Perimeter of a Rectangle

The perimeter of a rectangle can be found easily by measuring the four sides and adding them up.

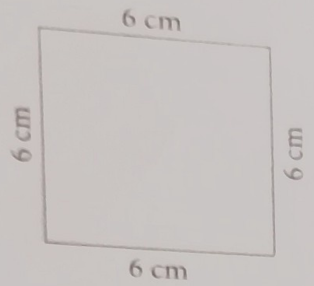
Let us find the perimeter of the rectangle given below.



Perimeter of rectangle

= 15 cm + 8 cm + 15 cm + 8 cm

= 46 cm



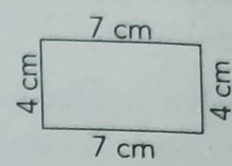
Perimeter of a Square

Let us consider a square. A square is a closed figure.
Its perimeter is obtained by adding the length of its four sides.
Let us find the perimeter of the given square.

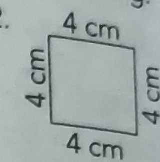
Let's Try! 2

Find the perimeter of the following.

1.



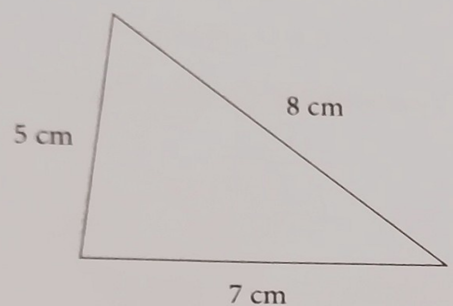
2.



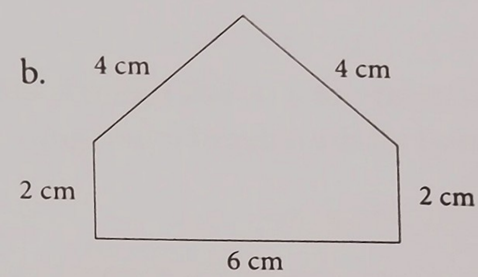
Perimeter of square = 6 cm + 6 cm + 6 cm + 6 cm
= 24 cm

Example 1: Find the perimeter of the following shapes.

a.



b.

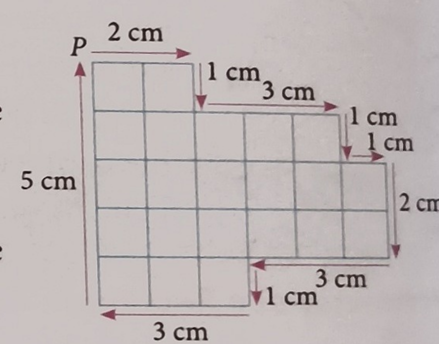


Solution: Perimeter = Length of the boundary = Sum of the sides

a. Perimeter = 8 cm + 7 cm + 5 cm = 20 cm

b. Perimeter = 4 cm + 2 cm + 6 cm + 2 cm + 4 cm = 18 cm

Perimeter of Some Other Shapes



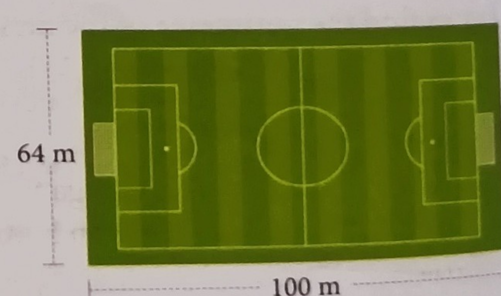
How will you find the perimeter of a figure such as the one given alongside?

The boundary is made up of line segments. The perimeter is the sum of all the lengths of the line segments that make up the boundary.

If each square is of side 1 cm, then the perimeter
= 2 cm + 1 cm + 3 cm + 1 cm + 1 cm + 2 cm + 3 cm + 1 cm + 3 cm + 5 cm
= 22 cm.

Example 2: A football field is 64 m wide and 100 m long. What is its perimeter?

(Note: Figure not drawn to scale)



Solution: Perimeter of the football field
= 100 m + 64 m + 100 m + 64 m
= 328 m

Example 3: A farmer has a square piece of land. Each side measures 34 m. He wants to put a fence around his land. What length of wire should he buy for the fence?

Solution: Perimeter of the land = 34 m + 34 m + 34 m + 34 m

= 136 m

The farmer has to buy 136 m of wire for the fence.

Example 4: The perimeter of the given figure is 16 cm. Find the missing length.

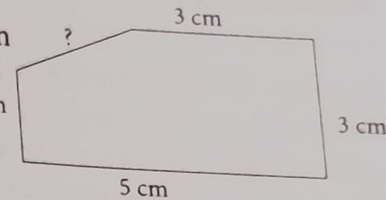
Solution: Perimeter = 16 cm

Sum of the given sides = 3 cm + 3 cm + 5 cm + 2 cm = 13 cm

Missing length = Perimeter – Sum of the given sides

= 16 cm – 13 cm = 3 cm

So, the length of the fifth side is 3 cm.

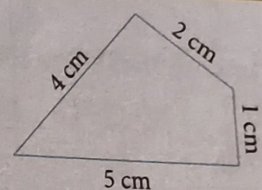


Go to MyInspire for

Exercise 14A

Go to MyInspire and attempt questions 1 to 3.

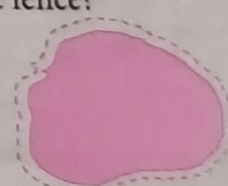
4. Find the perimeter of the given figure.

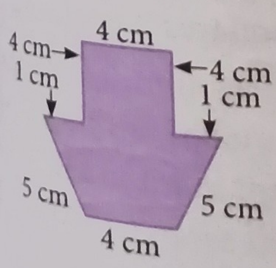


Application Based Questions

5. The length of a rectangular postcard is 10 cm and its breadth is 7 cm. Find the perimeter of the postcard.
6. The breadth of a rectangle is 20 cm shorter than the length. The length measures 52 cm. Find the perimeter of the rectangle.
7. The length of a rectangular field is 120 m and its width is 80 m. Ritu ran once around the field along the boundary. Find the distance Ritu ran.

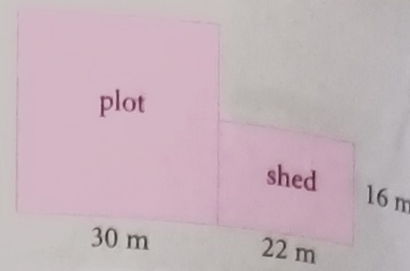
8. Mr Rangan wants to fence his garden. The garden is in the shape of a square with each side measuring 21 m. What length of wire he should purchase for the fence?
9. Vivek rode his bicycle two times around the pond. The perimeter of the pond is 1 km 500 m. What distance did Vivek ride?





10. Ganga wants to paste a ribbon along the edges of the shape given here. What length of ribbon would she need? *(Note: Figure not drawn to scale)*

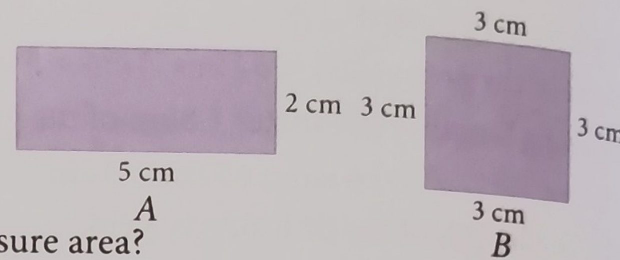
11. Ramu has a square plot of land of side 30 m and a rectangular shed of length 22 m and breadth 16 m next to the plot. He wants to fence both the plot and the shed. What length of barbed wire would he need?
(Note: Figure not drawn to scale)



AREA OF CLOSED FIGURES

Consider the given figures.

These are closed figures enclosing an extent of surface. The shaded surface is the region enclosed and is known as the **area** of the figure.



Which figure has greater area? How do we measure area?

Area cannot be measured using a ruler. We need a square-grid paper with 1-cm squares. The shape is traced out on the square-grid paper and the number of squares within the boundary is found. The number of 1-cm squares gives the area of the shape. The unit for area in this case is '**square centimetres**' or '**sq. cm**' in short.

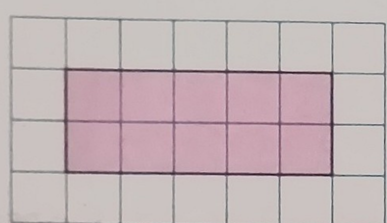


Figure *A*

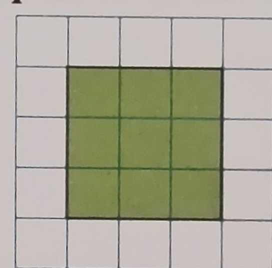


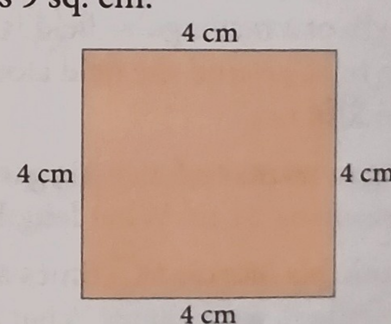
Figure *B*

We count the number of squares in each figure.

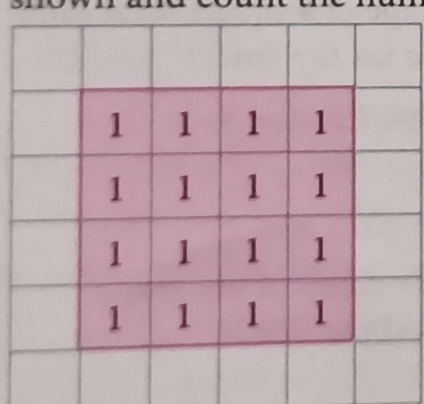
Figure A encloses 10 squares. So, the area of figure *A* is 10 sq. cm.

Figure B encloses 9 squares. So, the area of figure *B* is 9 sq. cm.

Example 5: Find the area of the given square.



Solution: We trace the square on a square grid as shown and count the number of squares enclosed.



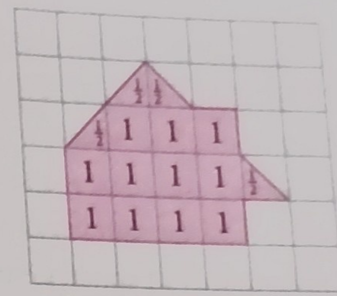
Area of the square is the number of 1-cm squares it encloses.

The side of the square measures 4 cm.

There are four rows of four 4-cm squares.

The area is (4 + 4 + 4 + 4) sq. cm = 16 sq. cm.

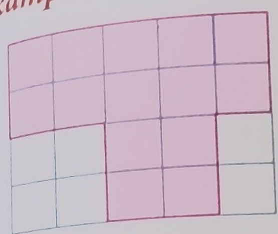
Example 6: Find the area of the given figure.
Take each square to have side 1 unit.



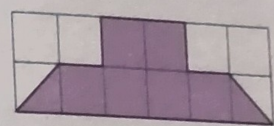
Solution: Each square is of area 1 sq. unit.
The number of complete squares within the boundary = 11
The number of half squares = 4
4 half squares $= \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{4}{2} = 2$ complete squares
The total number of squares covered = 11 + 2 = 13 squares
So, the area of the given shape 13 sq. units.

Example 7: Find the area of the following shapes. Take each square to have side 1 unit.

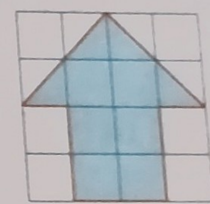
a.



b.



c.



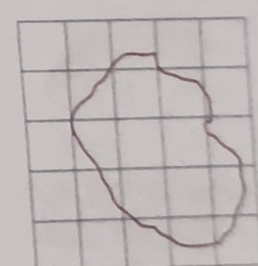
Solution:

a. Number of squares enclosed = 14.
Area = 14 sq. units.

b. The given shape covers 6 complete squares and 2 half squares.
2 half squares = 1 complete square.
Area = 6 + 1 = 7 sq. units.

c. The given shape covers 6 complete squares and 4 half squares.
4 half squares = 2 complete squares.
Area = 6 + 2 = 8 sq. units.

Area of Irregular Figures



The given figure is not made up of straight lines. So, we can only find its approximate area.

In the given figure, we notice that some of the squares inside the outline are full, some are less than half, some are more than half, and some may be exactly half.

We find the approximate area of the given figure as follows.

Step 1: Count the whole squares. The area of each complete square is 1 sq. unit.
Number of whole squares = 4
Area for whole squares = 4 sq. units.

Step 2: Count the squares that are more than half. The area of the square which is more than half the square is counted as 1 sq. unit.
Number of more than half squares = 5
Area of five more than half squares = 5 sq. units.

Step 3: Count the squares that are half. The area which is half is counted as $\frac{1}{2}$ sq. unit.
Number of half squares = 1

Area of one half square = $\frac{1}{2}$ sq. unit.

We ignore the squares that are less than half.

Step 4: Add the number of squares from Steps 1, 2, and 3.

Total number of squares enclosed by the figure $= 4 + 5 + \frac{1}{2}$

$= 9\frac{1}{2}$

The approximate area of figure = 4 sq. units + 5 sq. units + $\frac{1}{2}$ sq. unit = $9\frac{1}{2}$ sq. units.

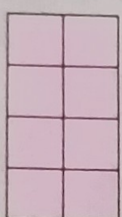
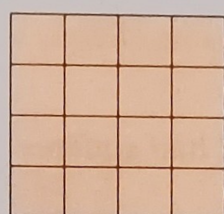
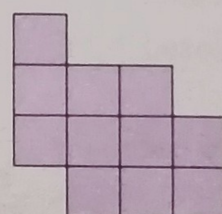
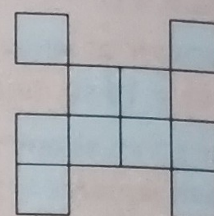
Go to MyInspire for

CREATING

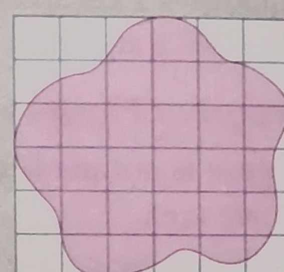
Exercise 14B

Go to MyInspire and attempt questions 1 and 2.

3. Find the area of the each figure. Each square is of area 1 sq. unit.

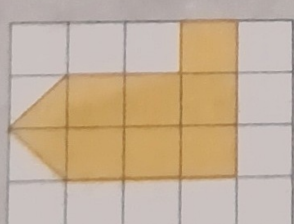
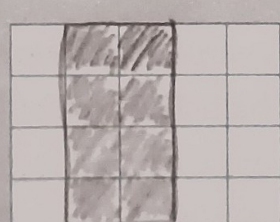
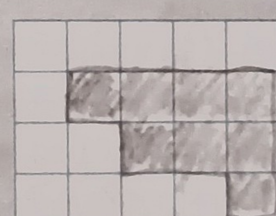
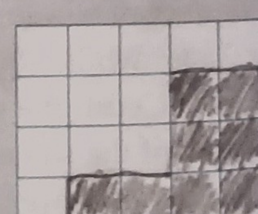
a. 
b. 
c. 
d. 

4. Find the approximate area of each shape. Each square is of area 1 sq. unit.

a. 
b. 
c. 

Application Based Questions

5. Make a different shape of area 8 sq. units on each grid.
You can use half and full squares. One has been done for you.

a. 
b. 
c. 
d. 

Let's Review

CENTURY SKILLS

Come here after you have completed the chapter. Colour 1 to 5 🙂 to show how well you understand a topic. (5 🙂 = Very well; 1 🙂 = Not at all)

Perimeter	🙂 🙂 🙂 🙂 🙂
Area of Closed Figures	🙂 🙂 🙂 🙂 🙂

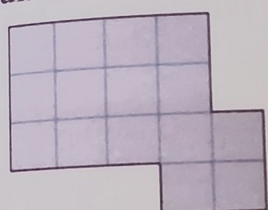
Revision Exercises

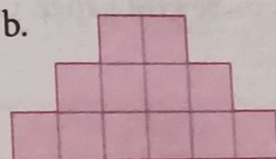
Go to MyInspire for

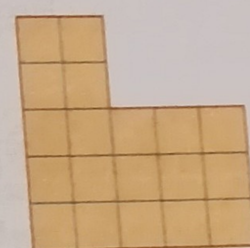
BLOOM'S REVISED TAXONOMY

Start Up REMEMBERING UNDERSTANDING

1. **Find the perimeter and area of the following figures. Take each square to have side 1 unit.**

a. 

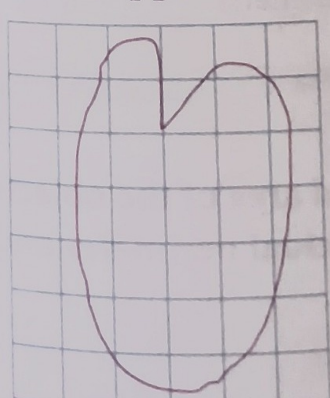
b. 

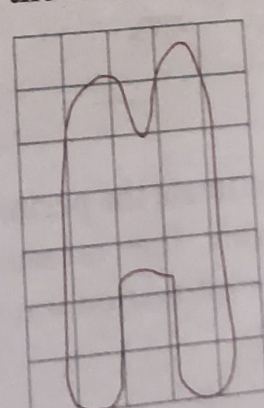
c. 

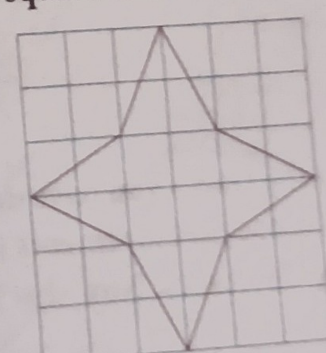
2. **Complete the table.**

	Shape	Dimensions	Perimeter
a.	Rectangle	length = 25 cm, breadth = 17 cm	
b.	Square	side = 41 cm	
c.	Rectangle	length = 15 m, breadth = 11 m	
d.	Square	side = 12 cm	

3. **Find the approximate area of the following. Take each square to have side 1 unit.**

a. 

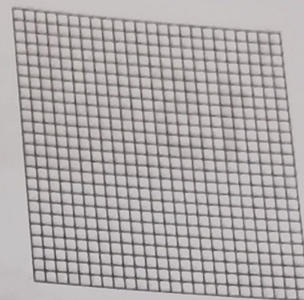
b. 

c. 

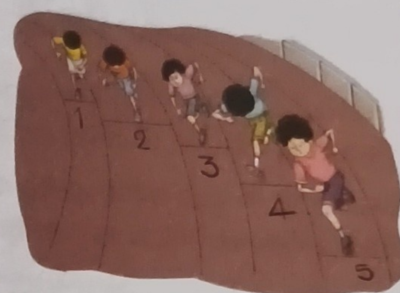
Name of the objects	Estimated measurements	Actual measurements
	Perimeter ______ Area ______	Perimeter ______ Area ______
	Perimeter ______ Area ______	Perimeter ______ Area ______
	Perimeter ______ Area ______	Perimeter ______ Area ______

HOTS Questions

1. How many different rectangles can have an area of 24 square units? You can use the following grid to find the answer.
   Note: Each square is of area 1 sq. unit.

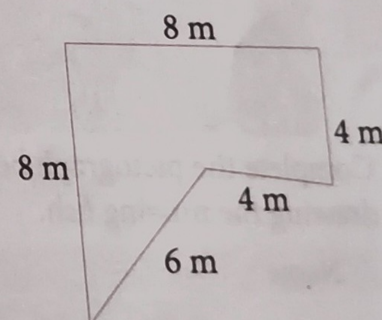


2. You must have seen the athletes running on the tracks.
   Why do you think all the athletes don't start from the same position on the tracks?



Everyday Maths

The idea of area and perimeter is used in daily life by those who work in the fields of architecture, fashion, interior design, engineering, aeronautical and graphical design, and farming. Even at home, we use the concept of area and perimeter most frequently. Razia has a garden which has not been maintained for some time. She wants to put a fence all around it.



1. Find the length of wire need for the fence.
2. If the cost of one metre wire is ₹18, find the total cost of wire needed for the fencing.
3. She also wants to plant trees along the boundary with gaps of 2 m between them. Find the number of trees she would plant.